Unit I Question Bank

Unit I: Boolean Algebra and Simplification Techniques

Boolean Algebra: Basic theorems and properties of Boolean algebra, DeMorgan's rules.

Q1: Simplify following logic function using Boolean algebra

$$f = A\overline{B}C + B\overline{C}D + \overline{A}BC$$

Q2: A staircase light is controlled by two switches, one at the top of the stairs and another at the bottom of stairs.

a) Make a truth table for this system
b) Write the logic equation in SOP form
c) Realise the circuit using AND-OR gates
d) Realise the circuit using NAND gates only

Signed Binary number representation: Signed Magnitude, 1's complement and 2's complement representation.

Q1: Represent +27, - 62 in

a) Signed magnitude
b) 1's complement
c) 2's complement

Q2: Represent +87, - 43 in

a) Signed magnitude
b) 1's complement
c) 2's complement

Simplification Techniques: Sum of product and Product of sum form,

Minimization of SOP and POS using K-map.

Q1: Convert following expression to canonical SOP form and minimize it using K-Map

Y = AB + AC + BCD

Q2: Convert following expression to canonical POS form and minimize it using K-Map

Y = (A+C+D)(B+C+D)(B+D)

Minimization of Boolean function using K-map (up to 4 variables).

Q1. Minimize the following function in SOP form using K-Map:

$f(A, B, C, D) = \sum m\ (0, 4, 5, 10, 11, 13, 15)$

Q2. Minimize the following function in POS form using K-Map:

$F1(A, B, C, D) = \prod M\ (0, 1, 2, 3, 11, 12, 14, 15)$

Q3. Minimize the following function in SOP form using K-Map:

$F1(A, B, C, D) = \sum m\ (0, 4, 5, 10, 11, 13, 15) + d(6, 14)$

Q4. Minimize the following function in POS form using K-Map:

$F1(A, B, C, D) = \prod M\ (1, 2, 8, 9, 10, 11, 14)\ .\ d(7, 15)$